The sixth-power Berndt integral
MathDB 361364

Research note

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Task 1. Prove the exact evaluation

Put $G = \Gamma(1/4)$. We prove Xu–Zhao's Conjecture 5.5 [1, p. 24], namely

$$\begin{aligned} I := \int_0^\infty \frac{x^9\,dx}{(\cos x + \cosh x)^6} = &-\frac{63}{5\cdot 2^{10}} + \frac{1071G^8}{5^2\cdot 2^{13}\pi^2} - \frac{21G^8}{2^{12}\pi^3} + \frac{63G^8}{2^{16}\pi^4} \\ &- \frac{21G^{16}}{5^3\cdot 2^{13}\pi^4} + \frac{3G^{16}}{5\cdot 2^{13}\pi^5} - \frac{161G^{16}}{5\cdot 2^{19}\pi^6} + \frac{21G^{16}}{2^{21}\pi^8} \\ &+ \frac{G^{24}}{5^2\cdot 2^{19}\pi^8} - \frac{G^{24}}{3\cdot 2^{20}\pi^9} + \frac{69G^{24}}{5\cdot 2^{25}\pi^{10}} \\ &- \frac{21G^{24}}{5\cdot 2^{24}\pi^{11}} + \frac{63G^{24}}{5\cdot 2^{27}\pi^{12}} - \frac{17G^{32}}{3\cdot 5^2\cdot 2^{31}\pi^{14}} \\ &+ \frac{13G^{32}}{5\cdot 2^{34}\pi^{16}} + \frac{3G^{40}}{5^2\cdot 2^{40}\pi^{20}}. \end{aligned} \tag{1}$$

The source's general structural theorem assumes $p \geq \lfloor m/2 \rfloor$ for a numerator x^{4p+1} and denominator power m. The present case $(p, m) = (2, 6)$ is outside that assumption. The missing ingredient is the elementary Lambert-series formula for S_0 below.

1. A convergent residue formula

Let

$$F(z) = \frac{z^9}{(\cos z + \cosh z)^6}, \qquad a = \frac{1+i}{2}, \quad b = \frac{1-i}{2}.$$

The factorization $\cos z + \cosh z = 2\cosh(az)\cosh(bz)$ shows that the only poles in the first quadrant are

$$z_n = (1+i)t_n, \qquad t_n = \pi(n + \tfrac{1}{2}), \quad n \geq 0,$$

each of order six. Integrate on the positively oriented triangle with vertices $0, R, iR$, where $R = 2\pi N$ and N is a positive integer. On its oblique side $x + y = R$, we have

$$|\cosh(az)| = \cosh((x-y)/2) \geq 1, \qquad |\cosh(bz)| \geq \sinh(R/2).$$

The modulus of the integral on that side is therefore at most $\sqrt{2}R^{10}/(64\sinh^6(R/2))$, which tends to zero. Since $F(iy) = iF(y)$, the two axial sides together give $2\int_0^R F(x)\,dx$. The residue theorem consequently yields

$$I = \pi i \sum_{n\geq 0} \operatorname{Res}_{z=z_n} F(z). \tag{2}$$

The original integral converges at zero and infinity. The residue series below is absolutely convergent, because each term is a finite sum of polynomials in t_n times derivatives of $\operatorname{sech}^6 t_n$.

Write $z = z_n + w$. Then

$$F(z_n + w) = -\frac{(z_n + w)^9}{64 \sinh^6(aw) \cosh^6(t_n + bw)}, \qquad \left(\frac{v}{\sinh v}\right)^6 = 1 - v^2 + \frac{8}{15}v^4 + O(v^6).$$

Extracting the coefficient of w^{-1} gives

$$\operatorname{Res}_{z=z_n} F(z) = \frac{i}{2} \sum_{h=0}^{2} \gamma_h \sum_{\ell=0}^{5-2h} \binom{9}{\ell} \frac{t_n^{9-\ell}}{r!} (\operatorname{sech}^6)^{(r)}(t_n), \quad r = 5 - 2h - \ell, \tag{3}$$

where $(\gamma_0, \gamma_1, \gamma_2) = (1, 1, 8/15)$. For example, if $(A_0, A_1, A_2) = (1, -1, 8/15)$, the phase of the term indexed by h, ℓ, r is

$$-\frac{i}{8} A_h a^{2h} (1 + i)^{9-\ell} b^r = \frac{i}{2} \gamma_h, \qquad \ell + r + 2h = 5.$$

This checks both the sign and all powers of two in (3).

Define, for even nonnegative m and $y > 0$,

$$S_m(y) = \sum_{n \geq 0} (n + \tfrac{1}{2})^m \operatorname{sech}^2((n + \tfrac{1}{2})y).$$

The elementary differential identity

$$\operatorname{sech}^6 t = \frac{1}{120} \left(\frac{d^4}{dt^4} - 20 \frac{d^2}{dt^2} + 64 \right) \operatorname{sech}^2 t$$

and (2)–(3) now imply the finite formula

$$I = -\frac{\pi}{240} \sum_{h=0}^{2} \gamma_h \sum_{\ell=0}^{5-2h} \binom{9}{\ell} \frac{\pi^{9-\ell}}{r!} \sum_{j \in \{0,2,4\}} c_j S_{4+2h-j}^{(r+j)}(\pi), \quad r = 5 - 2h - \ell, \tag{4}$$

where $(c_0, c_2, c_4) = (64, -20, 1)$. Only S_0, S_2, S_4, S_6, S_8 occur. Differentiation under each sum is justified by locally uniform exponential convergence for $y > 0$.

2. The five sums as Lambert series

For $|q| < 1$ and positive odd s, set

$$L_s(q) = \sum_{k \geq 1} \frac{k^s q^k}{1 - q^k}, \qquad D = q \frac{d}{dq}.$$

In the next identities $q = e^{-y}$. For $m \in \{2, 4, 6, 8\}$,

$$S_m(y) = 2^{2-m} D \left\{ L_{m-1}(q) - (2 + 2^{m-1}) L_{m-1}(q^2) + 2^m L_{m-1}(q^4) \right\}. \tag{5}$$

Indeed, $\operatorname{sech}^2(ky/2) = 4q^k/(1 + q^k)^2$, and

$$\sum_{\substack{k \geq 1 \\ k \text{ odd}}} \frac{k^{m-1} q^k}{1 + q^k} = L_{m-1}(q) - (2 + 2^{m-1}) L_{m-1}(q^2) + 2^m L_{m-1}(q^4).$$

Differentiating proves (5).

For the remaining case, interchange two absolutely convergent sums:

$$\frac{1}{4}S_0(y) = \sum_{\substack{k\geq 1\\ k \text{ odd}}} \sum_{v\geq 1} (-1)^{v-1} v q^{kv} = \sum_{v\geq 1} \frac{(-1)^{v-1} v q^v}{1-q^{2v}}$$
$$= L_1(q) - 5L_1(q^2) + 4L_1(q^4).$$

Thus

$$S_0(y) = 4\{L_1(q) - 5L_1(q^2) + 4L_1(q^4)\}. \tag{6}$$

This uses a weight-two Lambert series instead of a negative-index series, and is valid for every $y > 0$.

3. Exact modular values and derivatives

Use the standard normalization $q = e^{2\pi i\tau}$ for Eisenstein series. In this normalization,

$$L_1 = \frac{1-E_2}{24}, \qquad L_3 = \frac{E_4-1}{240}, \qquad L_5 = \frac{1-E_6}{504}, \qquad L_7 = \frac{E_4^2-1}{480}.$$

Ramanujan's differential identities [2, Section 7] are

$$DE_2 = \frac{E_2^2 - E_4}{12}, \qquad DE_4 = \frac{E_2E_4 - E_6}{3}, \qquad DE_6 = \frac{E_2E_6 - E_4^2}{2}. \tag{7}$$

They give every derivative needed in (4) by polynomial arithmetic over $\mathbb{Q}$.

Put

$$u = \theta_3(0 \mid i)^4 = \frac{G^4}{4\pi^3}.$$

The last equality is the classical elliptic-integral value also recorded in [1, equation (2.3)]. The three evaluation points in (5)–(6) give

τ	$E_2(\tau)$	$E_4(\tau)$	$E_6(\tau)$
$i/2$	$6/\pi - 3u/2$	$33u^2/4$	$-189u^3/8$
i	$3/\pi$	$3u^2/4$	0
$2i$	$3/(2\pi) + 3u/8$	$33u^2/64$	$189u^3/512$

(8)

For completeness these values follow from the classical theta identities as follows. At i the fourth powers of θ_2 and θ_4 are both $u/2$. Theta duplication [3] gives at $2i$

$$(\theta_2^4, \theta_3^4, \theta_4^4) = u\left(\frac{3-2\sqrt{2}}{8}, \frac{3+2\sqrt{2}}{8}, \frac{1}{\sqrt{2}}\right).$$

Substitute into

$$E_4 = \tfrac{1}{2}(\theta_2^8 + \theta_3^8 + \theta_4^8), \qquad E_6 = \tfrac{1}{2}(\theta_2^4 + \theta_3^4)(\theta_3^4 + \theta_4^4)(\theta_4^4 - \theta_2^4).$$

For weight two use

$$2E_2(2\tau) - E_2(\tau) = \tfrac{1}{2}(\theta_3^4 + \theta_4^4), \qquad E_2(-1/\tau) = \tau^2 E_2(\tau) + \frac{6\tau}{\pi i}.$$

The latter gives $E_2(i) = 3/\pi$. These formulas and the modular transformations $E_w(-1/\tau) = \tau^w E_w(\tau)$ for $w = 4, 6$ give all of (8).

Here are explicit instructions for the finite evaluation, including the chain factors. Introduce variables e_2, e_4, e_6 and the derivation

$$\mathcal{D} = \frac{e_2^2 - e_4}{12}\partial_{e_2} + \frac{e_2 e_4 - e_6}{3}\partial_{e_4} + \frac{e_2 e_6 - e_4^2}{2}\partial_{e_6}.$$

Let $\mathcal{L}_w = (1 - e_2)/24, (e_4 - 1)/240, (1 - e_6)/504, (e_4^2 - 1)/480$ for $w = 2, 4, 6, 8$, respectively. Let V_a denote substitution from row $\tau = ai/2$ of (8), for $a = 1, 2, 4$. For $d \geq 0$,

$$S_0^{(d)}(\pi) = 4(-1)^d \sum_{a=1,2,4} b_a a^d V_a(\mathcal{D}^d \mathcal{L}_2), \qquad (b_1, b_2, b_4) = (1, -5, 4), \tag{9}$$

$$S_m^{(d)}(\pi) = 2^{2-m}(-1)^d \sum_{a=1,2,4} b_{m,a} a^{d+1} V_a(\mathcal{D}^{d+1} \mathcal{L}_m), \qquad m = 2, 4, 6, 8, \tag{10}$$

where $(b_{m,1}, b_{m,2}, b_{m,4}) = (1, -2 - 2^{m-1}, 2^m)$. Here the factor $(-1)^d$ comes from $d/dy = -D$, and each factor a comes from differentiating q^a. Differentiation is performed *before* substituting the special values.

4. The resulting polynomial

Substitution of (9)–(10) into (4), followed by rational polynomial arithmetic, gives

$$\begin{aligned} I = & -\frac{63}{5120} + u^2\left(\frac{1071\pi^4}{12800} - \frac{21\pi^3}{256} + \frac{63\pi^2}{4096}\right) \\ & + u^4\left(-\frac{21\pi^8}{4000} + \frac{3\pi^7}{160} - \frac{161\pi^6}{10240} + \frac{21\pi^4}{8192}\right) \\ & + u^6\left(\frac{\pi^{10}}{3200} - \frac{\pi^9}{768} + \frac{69\pi^8}{40960} - \frac{21\pi^7}{20480} + \frac{63\pi^6}{163840}\right) \\ & + u^8\left(-\frac{17\pi^{10}}{2457600} + \frac{13\pi^8}{1310720}\right) + \frac{3\pi^{10}u^{10}}{26214400}. \end{aligned}$$

Replacing u by $G^4/(4\pi^3)$ is precisely (1). The accompanying exact checker implements the displayed derivation and substitutions and verifies zero difference from every one of the sixteen target terms. Its certificate records the nine grouped contributions indexed by (h, j) in (4), so the arithmetic can also be checked group by group. No numerical recognition of constants or unproved convergence assertion is used. □

Task 2. Remove the low-degree restriction

Extend this argument to $\int_0^\infty x^{4p+1}(\cos x + \cosh x)^{-m}\,dx$ when $p < \lfloor m/2 \rfloor$. Determine which negative-index sums produced by the residue calculation can be replaced by positive-weight Lambert series through interchange of divisor sums.

Task 3. Prove the fifth-moment pattern

Apply the resulting machinery to Conjecture 5.9 of [1], concerning all denominator powers with numerator x^5. Explain the exact allowed powers of $\Gamma(1/4)$ and π, including the predicted missing coefficient, rather than merely producing individual evaluations.

Task 4. Evaluate at other modular parameters

Replace the equal cosine and hyperbolic-cosine scales by a ratio tied to a general elliptic modulus. Identify analogous exact evaluations at other imaginary quadratic parameters and determine the necessary periods and algebraic coefficients.

Task 5. Find recurrences for exact evaluation

Develop recurrences in numerator degree and denominator power for the residue polynomials and their modular expressions. Measure the growth of intermediate expressions and identify a basis that makes the final cancellations visible before expansion.

References

[1] C. Xu and J. Zhao, *General Berndt-Type Integrals and Series Associated with Jacobi Elliptic Functions*, arXiv:2401.01385v2 (2024).

[2] S. Ramanujan, *On certain arithmetical functions*, Trans. Cambridge Philos. Soc. **22** (1916), 159–184, Section 7; online transcription.

[3] NIST Digital Library of Mathematical Functions, Section 20.7, theta identities and transformations, and Section 23.8, elliptic-function series.

Verification and status. Run `verify_evaluation.py` with Python, SymPy, and mpmath. Exact algebra verifies the finite certificate. The separate numerical checks are diagnostics only. The full current primary source retains Conjecture 5.5; dated source and claim audits accompany this note. The authoring agent performed the proof review, without independent peer review.